Class XII Maths Sample Paper – Set 3

Time Allowed: 30 Minutes.

General Instructions:

1. This Question paper contains - five sections A, Each section is compulsory. However, there are internal choices in some questions.

2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

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Section A		
Q. No.	**Questions**	**Marks**
1.	The sum of order and degree of the differential equation $(\frac{dy}{dx})^2 + 3y(\frac{d^2y}{dx^2}) = 0$ is (a) 1 (b) 2	1

	(c) 3 (d) 4	
2.	The maximum slope of curve $y = -x^3 + 3x^2 + 9x - 27$ is (a) 0 (b) 12 (c) 16 (d) 32	1
3.	If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$, then $P(\frac{A}{B})$ equal (a) $\frac{1}{9}$ (b) $\frac{2}{9}$ (c) $\frac{3}{9}$ (d) $\frac{4}{9}$	1
4.	The domain of $f(x) = \sin\text{-}1\ (-x^2)$is (a) (-1, 1)	1

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	(b) [-1, 1] (c) (-1, 1] (d) [-1, 1)	
5.	If $y = 3e^{2x} + 2e^{3x}$ then the value of $y_2 - 5y_1 + 6y$ is (a) 1 (b) 0 (c) 2 (d) 3	1
6.	If a leap year is selected at random, then what is the chance that it will contain 53 Tuesday? (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{7}$	1
7.	The function $f(x) = x^2 + 2$ is (a) one-one	1

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	(b) not one-one (c) one-one and onto (d) None of these (b) not one-one	
8.	If P(A) = $\frac{1}{2}$, P(B) =0, then P($\frac{A}{B}$) is (a) zero (b) 2 (c) not defined (d) 11	1
9.	If A is a matrix of order 3 × 3 such that |A| = 4, then |A-1| is equal to (a) 4 (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) 2	1
10.	The particular solution of the differential equation $\frac{dy}{dx}$ = y tan x at y = 1, x = 0 is	1

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	(a) $y = \cos x$ (b) $y = \sec x$ (c) $y \sin x = 6$ (d) $y = \tan x$	
11.	If A and B are two independent events with $P(A) = \frac{3}{5}$ and $P(B) = \frac{4}{9}$, then $P(A' \cap B')$ equal to (a) $\frac{4}{15}$ (b) $\frac{8}{45}$ (c) $\frac{1}{3}$ (d) $\frac{2}{9}$	1
12.	The radius of the base of a cone is increasing at the rate of 3 cm/min and the altitude is decreasing at the rate of 4 cm/min. The rate of change of lateral surface when the radius is 7 cm and altitude 24 cm, is (a) 54n cm^2/min (b) 7 n cm^2/min (c) 27 cm^2/min (d) None of the above	1

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13.	The degree of the differential equation $(\frac{dy}{dx})^5 = 2x^2(\frac{d^2y}{dx^2})^4 = 0$ is (a) 1 (b) 2 (c) 3 (d) 4	1
14.	If y = tan x $\frac{1}{3}$ log x – $\frac{2}{x}$, then $\frac{dy}{dx}$ is equal to (a) $\sec x - \frac{1}{6x} + \frac{2}{x^2}$ (b) $\sec^2 x - \frac{1}{3x} + \frac{2}{x^2}$ (c) sec x – 3x + x^2 (d) None of these	1
15.	The angle between the lines through the points (4, 7, 8), (2, 3, 4) and (-1, -2, 1), (1, 2, 5) is (a) 0 (b) $\frac{\pi}{2}$ (c) $\frac{2\pi}{4}$	1

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	(d) $\frac{-\pi}{6}$	
16.	The corner points of the feasible region determined by the following system of linear inequalities $2x + y \leq 10$, $x + 3y \leq 15$, $x, y \geq 0$ are (0, 0) (5, 0) (3, 4) and (0, 5). Let $Z = px + qy$, where $p, q > 0$. Condition on p and q, so that the maximum of Z occurs at both (3, 4) and (0, 5), is (a) $p = q$ (b) $p = 2q$ (c) $p = 3q$ (d) $q = 3p$	1
17.	If $\lvert\vec{a}\rvert = 20$, $\lvert\vec{b}\rvert = 4$ and $\vec{a}.\vec{b} = 24$, then $\lvert\vec{a}\times\vec{b}\rvert$ is equal to (a) 30 (b) 32 (c) 0 (d) 24	1
18.	$\sin^{-1}(\cos \frac{3\pi}{5})$ is equal to (a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{5}$	1

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	(c) $\frac{-\pi}{10}$ (d) $\frac{-3\pi}{5}$	
19.	Assertion-Reason Based Questions In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices. (a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true Assertion (A) Let a relation R defined from set A = {1, 2, 5, 6} to A is R = {(1, 1), (1, 6), (6, 1)}, then R is symmetric relation. Reason (R) A relation R in set A is called symmetric, if (a, b) ∈ R ⇒ (b, a) ∈ R for every a, b ∈ A.	1
20.	Assertion-Reason Based Questions In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices. (a) Both A and R are true and R is the correct explanation of A	1

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	(b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true Assertion (A) The equation of the line joining A(1, 3) and B(0, 0) is given by y = 3x. Reason (R) The area of triangle with vertices (x_1, y_1) (x_2, y_3) and (x_3, y_3) in the form of determinant is $\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$.	

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